

LINUX FUNDAMENTALS FOR ODOO WORK

Linux OS & CLI Mastery

Essential terminal commands for Odoo developers

Merit Advisory

Odoo Full-Stack Developer Program

Beginner-friendly lecture materials

July 2026

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Agenda

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1 Section 1

Getting oriented in the terminal

1.1 pwd - Print Working Directory

- Stands for Print Working Directory and shows the full path of your current directory.
- Use it before running file commands so you know exactly where you are.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ pwd
/home/trainee
trainee@odoo-lab:~$ cd project
trainee@odoo-lab:~/project$ pwd
/home/trainee/project
trainee@odoo-lab:~/project$ echo "You are in: $(pwd)"
You are in: /home/trainee/project
```

1.2 cd - Go to your home directory

- The cd command changes the directory your shell is working in.
- Running cd with no path takes you back to your home directory.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project/addons$ cd
trainee@odoo-lab:~$ pwd
/home/trainee
trainee@odoo-lab:~$ cd ~/Downloads
trainee@odoo-lab:~/Downloads$ pwd
/home/trainee/Downloads
```

1.3 cd .. - Move up one directory

- Two dots mean the parent directory of your current location.
- Use `cd ..` when you need to move one level higher in the directory tree.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project/addons$ pwd
/home/trainee/project/addons
trainee@odoo-lab:~/project/addons$ cd ..
trainee@odoo-lab:~/project$ pwd
/home/trainee/project
trainee@odoo-lab:~/project$ cd ../../
trainee@odoo-lab:~$ pwd
/home/trainee
```

1.4 cd with absolute paths

- An absolute path starts with a slash and works from anywhere.
- Use absolute paths when you want commands to be unambiguous.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ cd /opt/odoo
trainee@odoo-lab:/opt/odoo$ pwd
/opt/odoo
trainee@odoo-lab:/opt/odoo$ cd /var/log
trainee@odoo-lab:/var/log$ pwd
/var/log
```

1.5 cd with relative paths

- A relative path starts from the directory you are currently in.
- Relative paths are short and useful while working inside a project.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ ls
addons  odoo.conf
trainee@odoo-lab:~/project$ cd addons
trainee@odoo-lab:~/project/addons$ cd ../custom_addons
trainee@odoo-lab:~/project/custom_addons$ pwd
/home/trainee/project/custom_addons
```


1.6 ls - List files

- The ls command shows files and directories in the current location.
- Use it after cd to confirm what is available before opening or editing files.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ ls
addons  custom_addons  odoo.conf  README.md
trainee@odoo-lab:~/project$ ls -lah
drwxr-xr-x  5 trainee trainee 4.0K Jul 25 09:10 .
drwxr-x--- 18 trainee trainee 4.0K Jul 25 08:01 ..
drwxr-xr-x  3 trainee trainee 4.0K Jul 25 09:10 addons
-rw-r--r--  1 trainee trainee 812 Jul 25 09:08 odoo.conf
```

1.7 ls -l - Long listing

- The -l option shows permissions, owner, group, size, and date.
- Use it when checking who owns a file or whether it is executable.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ ls
addons  custom_addons  odoo.conf  README.md
trainee@odoo-lab:~/project$ ls -lah
drwxr-xr-x  5 trainee trainee 4.0K Jul 25 09:10 .
drwxr-x--- 18 trainee trainee 4.0K Jul 25 08:01 ..
drwxr-xr-x  3 trainee trainee 4.0K Jul 25 09:10 addons
-rw-r--r--  1 trainee trainee 812 Jul 25 09:08 odoo.conf
```

1.8 ls -a -h - Hidden and readable sizes

- The -a option shows hidden files that start with a dot.
- The -h option makes sizes easier to read when used with -l.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ ls
addons  custom_addons  odoo.conf  README.md
trainee@odoo-lab:~/project$ ls -lah
drwxr-xr-x  5 trainee trainee 4.0K Jul 25 09:10 .
drwxr-x--- 18 trainee trainee 4.0K Jul 25 08:01 ..
drwxr-xr-x  3 trainee trainee 4.0K Jul 25 09:10 addons
-rw-r--r--  1 trainee trainee 812 Jul 25 09:08 odoo.conf
```

1.9 tree - View directory structure

- The tree command displays directories and files as a visual hierarchy.
- It helps beginners understand project layouts faster than repeated ls commands.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ tree -L 2
.
├── addons
│   └── wl_student
├── custom_addons
└── odoo.conf
```

2 Section 2

Working with files and directories

2.1 mkdir - Create directories

- The mkdir command creates a new directory.
- Use -p when parent directories may not exist yet.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ mkdir -p custom_addons/wl_student
trainee@odoo-lab:~/project$ touch custom_addons/wl_student/__init__.py
trainee@odoo-lab:~/project$ ls custom_addons/wl_student
__init__.py
```

2.2 touch - Create empty files

- The touch command creates an empty file if it does not exist.
- If the file already exists, touch updates its timestamp.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ mkdir -p custom_addons/wl_student
trainee@odoo-lab:~/project$ touch custom_addons/wl_student/__init__.py
trainee@odoo-lab:~/project$ ls custom_addons/wl_student
__init__.py
```

2.3 cp - Copy a file

- The cp command copies a file to a new path.
- Use it before editing important configuration files.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ cp odoo.conf odoo.conf.bak
trainee@odoo-lab:~/project$ ls odoo.conf*
odoo.conf  odoo.conf.bak
trainee@odoo-lab:~/project$ cp -r addons addons_copy
trainee@odoo-lab:~/project$ ls -d addons*
addons  addons_copy
```


2.4 cp -r - Copy directories

- The -r option copies directories recursively.
- Recursive copy includes files and directories inside the source directory.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ cp odoo.conf odoo.conf.bak
trainee@odoo-lab:~/project$ ls odoo.conf*
odoo.conf  odoo.conf.bak
trainee@odoo-lab:~/project$ cp -r addons addons_copy
trainee@odoo-lab:~/project$ ls -d addons*
addons  addons_copy
```

2.5 mv - Rename files

- The mv command can rename a file in the same directory.
- Renaming does not create a second copy.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ mkdir -p configs
trainee@odoo-lab:~/project$ mv odoo.conf.bak configs/odoo.conf.bak
trainee@odoo-lab:~/project$ ls configs
odoo.conf.bak
```

2.6 mv - Move files

- The mv command also moves files into another directory.
- Moving changes the file path but keeps the same file content.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ mkdir -p configs
trainee@odoo-lab:~/project$ mv odoo.conf.bak configs/odoo.conf.bak
trainee@odoo-lab:~/project$ ls configs
odoo.conf.bak
```

2.7 rm - Remove files

- The rm command deletes files.
- Deleted files usually do not go to a recycle bin on servers.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ rm odoo.conf.tmp
trainee@odoo-lab:~/project$ rm -r addons_copy
trainee@odoo-lab:~/project$ ls -d addons*
addons
```

2.8 rm -r - Remove directories carefully

- The -r option deletes directories and everything inside them.
- Use recursive deletion only after confirming the exact path.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ rm odoo.conf.tmp
trainee@odoo-lab:~/project$ rm -r addons_copy
trainee@odoo-lab:~/project$ ls -d addons*
addons
```

2.9 cat - Print file contents

- The cat command prints a whole file to the terminal.
- Use it for short files such as simple notes or small config snippets.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ cat odoo.conf | head -n 5
[options]
admin_passwd = strong-password
db_host = False
db_port = False
db_user = odoo
```

2.10 less and more - Page through files

- less opens a file one screen at a time.
- Use slash inside less to search within the file.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ cat odoo.conf | head -n 5
[options]
admin_passwd = strong-password
db_host = False
db_port = False
db_user = odoo
```

2.11 head and tail - Start or end of files

- head shows the beginning of a file.
- tail shows the end of a file, which is useful for recent logs.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ head -n 3 odoo.conf
[options]
admin_passwd = strong-password
db_host = False
trainee@odoo-lab:~/project$ tail -n 2 odoo.conf
logfile = /var/log/odoo/odoo.log
log_level = info
```


2.12 grep - Search text

- The grep command finds lines that contain matching text.
- Use -i for case-insensitive searches.

```
user@odoo-lab:~/project

trainee@odoo-lab:~/project$ grep -n "addons_path" odoo.conf
12:addons_path = /home/trainee/project/addons,/opt/odoo/addons
```

2.13 find - Locate files

- The find command searches directories by name, type, size, or time.
- Start with a specific directory so the search stays fast.

```
user@odoo-lab:~/project

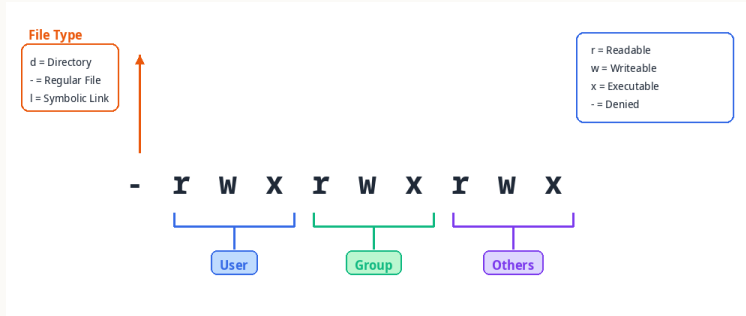
trainee@odoo-lab:~/project$ find . -name "*.py" | head
./addons/wl_student/__init__.py
./addons/wl_student/models/student.py
```

3 Section 3

Users, groups, and permissions

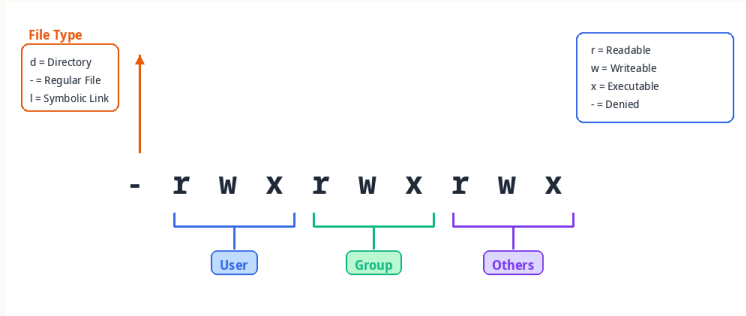
3.1 chmod numeric - Set permissions

- chmod changes read, write, and execute permissions.
- Numeric modes use 4 for read, 2 for write, and 1 for execute.



3.2 chmod symbolic - Adjust permissions

- Symbolic modes change permissions by user, group, or others.
- Use + to add a permission and - to remove one.



3.3 chown - Change owner

- chown changes which user owns a file or directory.
- The owner controls permissions marked for user access.

```
user@odoo-lab:~/project

trainee@odoo-lab:/opt$ sudo chown -R odoo:odoo /opt/odoo
trainee@odoo-lab:/opt$ ls -ld /opt/odoo
drwxr-xr-x 8 odoo odoo 4096 Jul 24 09:00 /opt/odoo
```

3.4 chgrp - Change group

- chgrp changes the group assigned to a file or directory.
- Group permissions help teams share access without using one account.

```
user@odoo-lab:~/project

trainee@odoo-lab:/opt$ sudo chown -R odoo:odoo /opt/odoo
trainee@odoo-lab:/opt$ ls -ld /opt/odoo
drwxr-xr-x 8 odoo odoo 4096 Jul 24 09:00 /opt/odoo
```

3.5 Users and groups overview

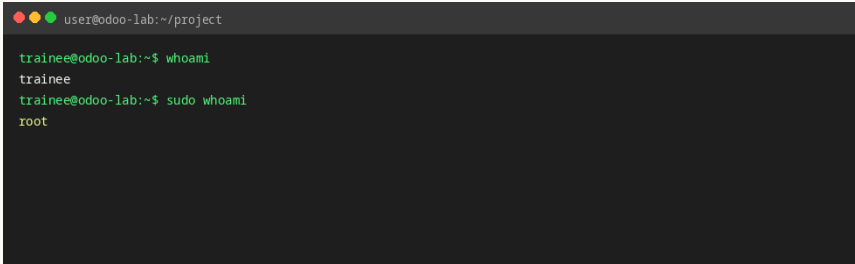
- A Linux user is an identity that owns processes and files.
- A group collects users so permissions can be shared.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ whoami
trainee
trainee@odoo-lab:~$ sudo whoami
root
```


3.6 sudo - Run with administrator rights

- sudo runs a command with elevated privileges.
- Use it only for commands that really need administrator access.



```
user@odoo-lab:~/project

trainee@odoo-lab:~$ whoami
trainee
trainee@odoo-lab:~$ sudo whoami
root
```

3.7 whoami and id - Check identity

- whoami prints the username of the current shell.
- id shows the user id, group id, and group memberships.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ whoami
trainee
trainee@odoo-lab:~$ sudo whoami
root
```

4 Section 4

Processes, services, logs, and editors

4.1 ps - List processes

- A process is a running program.
- ps shows processes and their process ids.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ sudo systemctl status postgresql
• postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/lib/systemd/system/postgresql.service; enabled)
   Active: active (exited) since Fri 2026-07-24 08:01:12 UTC
trainee@odoo-lab:~$ sudo systemctl start postgresql
trainee@odoo-lab:~$ sudo systemctl enable postgresql
```

4.2 top and htop - Watch system activity

- top shows live CPU, memory, and process activity.
- htop is an easier interactive version when installed.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ sudo systemctl status postgresql
• postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/lib/systemd/system/postgresql.service; enabled)
   Active: active (exited) since Fri 2026-07-24 08:01:12 UTC
trainee@odoo-lab:~$ sudo systemctl start postgresql
trainee@odoo-lab:~$ sudo systemctl enable postgresql
```

4.3 kill - Stop a process by id

- kill sends a signal to a process id.
- The default signal asks the process to stop cleanly.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ sudo systemctl status postgresql
• postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/lib/systemd/system/postgresql.service; enabled)
   Active: active (exited) since Fri 2026-07-24 08:01:12 UTC
trainee@odoo-lab:~$ sudo systemctl start postgresql
trainee@odoo-lab:~$ sudo systemctl enable postgresql
```

4.4 pkill - Stop processes by name

- pkill matches processes by name or pattern.
- It is faster than finding each process id manually.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ sudo systemctl status postgresql
• postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/lib/systemd/system/postgresql.service; enabled)
   Active: active (exited) since Fri 2026-07-24 08:01:12 UTC
trainee@odoo-lab:~$ sudo systemctl start postgresql
trainee@odoo-lab:~$ sudo systemctl enable postgresql
```

4.5 systemctl status - Check services

- systemctl manages services on systemd-based Linux systems.
- The status command shows whether a service is active and healthy.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ sudo systemctl status postgresql
• postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/lib/systemd/system/postgresql.service; enabled)
   Active: active (exited) since Fri 2026-07-24 08:01:12 UTC
trainee@odoo-lab:~$ sudo systemctl start postgresql
trainee@odoo-lab:~$ sudo systemctl enable postgresql
```


4.6 systemctl start and stop

- start launches a service managed by systemd.
- stop asks systemd to shut the service down.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ sudo systemctl status postgresql
• postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/lib/systemd/system/postgresql.service; enabled)
   Active: active (exited) since Fri 2026-07-24 08:01:12 UTC
trainee@odoo-lab:~$ sudo systemctl start postgresql
trainee@odoo-lab:~$ sudo systemctl enable postgresql
```

4.7 systemctl enable - Start on boot

- enable configures a service to start automatically after reboot.
- disable prevents automatic startup without deleting the service.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ sudo systemctl status postgresql
• postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/lib/systemd/system/postgresql.service; enabled)
   Active: active (exited) since Fri 2026-07-24 08:01:12 UTC
trainee@odoo-lab:~$ sudo systemctl start postgresql
trainee@odoo-lab:~$ sudo systemctl enable postgresql
```

4.8 journalctl - Read service logs

- journalctl reads logs collected by systemd.
- Use -u to focus on one service.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ sudo systemctl status postgresql
• postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/lib/systemd/system/postgresql.service; enabled)
   Active: active (exited) since Fri 2026-07-24 08:01:12 UTC
trainee@odoo-lab:~$ sudo systemctl start postgresql
trainee@odoo-lab:~$ sudo systemctl enable postgresql
```

4.9 nano - Beginner-friendly editor

- nano is a simple terminal text editor.

[Edit a note](#)

```
nano notes.txt
```

- The most important shortcuts are shown at the bottom of the screen.

[Edit Odoo config](#)

```
sudo nano /etc/odoo.conf
```

4.10 vim - Modal editor intro

- vim is a powerful editor with separate normal and insert modes.

Open a file

```
vim notes.txt
```

- Press i to insert text, Esc to return to normal mode, and :wq to save and quit.

Save and quit

```
:wq
```

5 Section 5

Networking, SSH, shell setup, and scheduling

5.1 ip - Inspect network addresses

- The ip command shows network interfaces, addresses, and routes.

Show addresses

```
ip addr
```

- Use it to confirm the server address before testing access.

Show default route

```
ip route
```

5.2 ping - Test basic reachability

- ping sends small network packets to test whether a host responds.
- It helps separate network reachability from application problems.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ ping -c 2 8.8.8.8
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=12.4 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=11.9 ms
--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss
```


5.3 ss - Check listening ports

- ss shows network sockets and listening ports.

Show listening services

```
sudo ss -tulpn
```

- Use it to confirm whether Odoo is listening on its expected port.

Check Odoo port

```
sudo ss -tulpn | grep 8069
```

5.4 SSH keys - Safer server login

- SSH keys let you log in without sending a password each time.
- A private key stays on your machine and a public key goes on the server.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ ssh-keygen -t ed25519 -C "odoo-admin"
Generating public/private ed25519 key pair.
Your identification has been saved in /home/trainee/.ssh/id_ed25519
trainee@odoo-lab:~$ ssh-copy-id ubuntu@server.example.com
Number of key(s) added: 1
```

5.5 SSH hardening basics

- SSH hardening reduces the chance of unauthorized server access.
- Common basics include key-based login, disabling root login, and limiting users.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ ssh-keygen -t ed25519 -C "odoo-admin"
Generating public/private ed25519 key pair.
Your identification has been saved in /home/trainee/.ssh/id_ed25519
trainee@odoo-lab:~$ ssh-copy-id ubuntu@server.example.com
Number of key(s) added: 1
```

5.6 Environment variables

- Environment variables are named values available to commands and scripts.

[Show all variables](#)

```
printenv
```

- They often hold settings such as paths, modes, or connection details.

[Set one for this shell](#)

```
export OD00_CONF=/etc/odoo.conf
```

5.7 .bashrc - Shell startup settings

- .bashrc runs when an interactive Bash shell starts.

Edit bashrc

```
nano ~/.bashrc
```

- Use it for aliases, prompts, and personal environment defaults.

Reload bashrc

```
source ~/.bashrc
```

5.8 cron - Schedule recurring commands

- cron runs commands on a time schedule.
- Each crontab line includes timing fields and a command.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ crontab -l
0 2 * * * /opt/odoo/scripts/backup.sh
trainee@odoo-lab:~$ crontab -e
# edit schedule, save, then verify with crontab -l
```

6 Section 6

Packages, resources, firewall, and Odoo operations

6.1 apt update - Refresh package lists

- apt update downloads the latest package index from configured repositories.
- It does not install upgrades by itself.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Reading package lists... Done
trainee@odoo-lab:~$ sudo apt install -y htop
Setting up htop (3.0.5-7build2) ...
```


6.2 apt install and remove

- apt install adds packages from configured repositories.
- apt remove uninstalls packages while usually leaving config files behind.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Reading package lists... Done
trainee@odoo-lab:~$ sudo apt install -y htop
Setting up htop (3.0.5-7build2) ...
```

6.3 df - Check filesystem space

- df shows free and used space on mounted filesystems.
- Use -h for human-readable units.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       98G   42G   52G   45% /
trainee@odoo-lab:~$ du -sh /opt/odoo
3.8G    /opt/odoo
```

6.4 du - Check directory size

- du shows how much space files and directories use.
- Use -s for a summary and -h for readable sizes.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       98G   42G   52G   45% /
trainee@odoo-lab:~$ du -sh /opt/odoo
3.8G    /opt/odoo
```

6.5 free - Check memory

- free shows system memory and swap usage.
- Use -h to make the numbers easier to read.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       98G   42G   52G   45% /
trainee@odoo-lab:~$ du -sh /opt/odoo
3.8G    /opt/odoo
```

6.6 ufw - Basic firewall management

- ufw is a simpler firewall tool used on many Ubuntu servers.

Show firewall status

```
sudo ufw status verbose
```

- Allow only the ports the server actually needs.

Allow SSH

```
sudo ufw allow OpenSSH
```

6.7 Production directories for Odoo

- Common Odoo deployments place application files under /opt/odoo.
- Configuration often lives in /etc and logs usually live under /var/log.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ sudo tail -n 20 /var/log/odoo/odoo.log
INFO mydb odoo.modules.loading: Modules loaded.
INFO mydb odoo.service.server: HTTP service (werkzeug) running
```

6.8 Odoo logs in production

- Odoo logs record startup messages, errors, warnings, and request details.
- Logs may be written to `/var/log/odoo` or collected by `journalctl`.

```
user@odoo-lab:~/project

trainee@odoo-lab:~$ sudo tail -n 20 /var/log/odoo/odoo.log
INFO mydb odoo.modules.loading: Modules loaded.
INFO mydb odoo.service.server: HTTP service (werkzeug) running
```

6.9 Safe troubleshooting checklist

- Observe the current state before making changes.

Capture service state

```
systemctl status odoo && journalctl -u odoo -n 50
```

- Change one thing at a time and record what changed.

Backup before editing

```
sudo cp /etc/odoo.conf /etc/odoo.conf.before-fix
```


NEXT STEP

Try the examples on your machine

Then open the next module deck

Merit Advisory

01 · Linux

